CLIPPER: Replayable Shortlisted Optimization for Repeated Spatial Coverage Planning

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Abstract

Municipal micromobility planning seeks coverage under geofenced exclusions, locks, spacing, or accounting-group caps. Workshops also compare alternatives as stakeholders revise these constraints. Each edit requires a new feasible plan; full-set greedy takes tens of seconds per alternative at city scale. We present Constraint-exact Low-latency Iterative Planning with Pooled Evaluation and Replay (**CLIPPER**), an optimizer over restricted candidate sets that separates recorded planning scenarios, bounded candidate generation, deterministic greedy selection with exact hard-constraint checks, and missed-gain auditing. CLIPPER uses cached singleton coverage to construct bounded candidate pools but evaluates current exact coverage marginals before every selection. Exact missed-gain audits require full-set scans and are reserved for offline or high-stakes checks; a conservative singleton-based bound is used online only to trigger pool expansion or an audit. Across Braunschweig, Munich, and Berlin, CLIPPER-F's complete-chain mean coverage stays within 0.245 percentage points of its same-policy full-set greedy control while achieving 13.6–28.9× mean rollout-time speedups. Under its coverage-prioritized policy, CLIPPER-A uses 9–15% of its same-policy control's rollout time with mean gaps of 1.82/0.12/0.27 percentage points. Together, these results establish CLIPPER as a replayable optimizer for rapid comparison of explicit city-scale planning states with exact feasibility checks under the evaluated weighted point-coverage model.

CCS Concepts

• **Information systems** → **Decision support systems**; • **Theory of computation** → *Submodular optimization and polymatroids*; • **Applied computing** → *Transportation.*

Keywords

spatial decision support, facility location, submodular coverage, auditability

ACM Reference Format:

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1 Motivation and contribution

Shared-micromobility parking-zone design seeks demand coverage under exclusions, retained sites, spacing, and area-level allocations. Such optimization also supports workshops: planners and stakeholders edit constraints, inspect coverage effects, and compare policy-feasible alternatives over common demand and candidates. Each edit requires reoptimization. Braunschweig grounds this setting: its municipal quality agreement permits riding and parking restrictions to be revised at any time, including temporary no-return areas for major events [11]. Full-set greedy takes tens of seconds per alternative at city scale; CLIPPER targets seconds-level feedback while retaining explicit constraints, replay records, and missed-gain auditing. Planning-support research treats such what-if exploration as an aid to deliberation [6, 10].

Maximal covering and monotone submodular optimization provide transparent siting models [4, 9]. Lazy, stochastic, and thresholded greedy methods reduce marginal-query cost [2, 7, 8]. Lazy evaluation retains the full candidate universe and prunes redundant queries; stochastic and thresholded variants trade queries for approximation under their stated constraint models. Dynamic and prediction-augmented methods instead maintain solutions across element-update sequences [1]. The closest dynamic method assumes a cardinality constraint and does not directly encode our combined locks, accounting-group caps, conflict classes, and network spacing. Prior micromobility work has studied geofenced parking and strategic facility placement [3, 12, 13]. CLIPPER (Constraint-exact Low-latency Iterative Planning with Pooled Evaluation and Replay) contributes a system-level execution contract rather than a new approximation primitive. It combines a replaceable bounded candidate generator, same-policy full-set controls, and distinct online screening and offline audit paths. Whereas dynamic-update methods maintain a solution across edits, CLIPPER independently recomputes each recorded state.

2 CLIPPER

Planning model. Let V be the feasible facility candidates and $\mathcal{D}$ weighted demand points with $w_d \geq 0$. Candidate e covers $C_e \subseteq \mathcal{D}$, giving the monotone submodular objective

$$f(S) = \sum_{d \in \mathcal{D}} w_d \, \mathbf{1}\left[d \in \bigcup_{e \in S} C_e \right]. \tag{1}$$

A recorded scenario Ω defines active candidates V_Ω, mandatory locks $L(\Omega)$, and a feasible family $\mathcal{I}(\Omega)$. Feasible sets contain every lock, exclude all unlocked candidates inside exclusion zones, obey

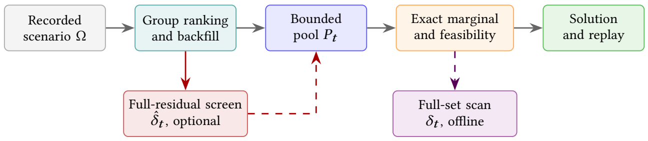


Figure 1: CLIPPER execution contract. Solid arrows show the reported optimization path; the optional full-residual screen can request expansion, while exact full-set scans support offline or high-stakes audits and are excluded from reported rollout time.

the global budget and accounting-group caps, take at most one site per conflict class, and satisfy minimum network spacing. Locks override exclusions; evaluated scenarios are lock-feasible. CLIPPER seeks to maximize $f(S)$; same-policy full-set greedy is an empirical reference, not an optimum. Figure 1 gives the execution contract.

Restricted selection with exact feasibility checks. Proposal groups partition the candidate rows. Within each group, CLIPPER ranks candidates once by singleton coverage $\mathrm{pot}(e) = f(\{e\})$, with stable tie-breaking. At round t, it filters invalid or already selected entries and backfills a bounded pool P_t. Starting from the locked facilities, the selector chooses

$$e_t \in \operatorname*{arg\,max}_{\substack{e \in P_t \\ S_{t-1} \cup \{e\} \in \mathcal{I}(\Omega)}} \Delta(e \mid S_{t-1}), \quad \Delta(e \mid S) = f(S \cup \{e\}) - f(S). \quad (2)$$

Thus shortlisting determines which candidates receive exact marginal evaluations, while the selector checks every active hard constraint before insertion. This guarantees feasibility, not optimality. Each proposal group maps to one accounting group. CLIPPER-F uses a fixed per-group width K under a policy that divides the residual facility budget equally among accounting groups. CLIPPER-A allocates a total per-round candidate budget B_{prop} toward groups with more feasible candidates and larger maximum singleton scores. Its coverage-prioritized policy relaxes candidate-count-weighted accounting caps by $\lambda = 1.5$ while preserving the global facility budget. We compare CLIPPER-A only with a full-set control under those same caps.

Feasibility follows by induction: $S_0 = L(\Omega) \in \mathcal{I}(\Omega)$, and every accepted insertion is checked against $\mathcal{I}(\Omega)$. In the evaluated instances, K-means partitions candidate rows; the implementation filters and deduplicates proposals within each group before selection. The singleton order is computed once. At every round, CLIPPER refilters candidates, backfills the pool, and recomputes exact current marginals to account for demand already covered. Recorded inputs and deterministic tie-breaking make the execution replayable.

Let $W_t = |P_t|$ after filtering and deduplication, T be the number of insertions, and $c = \max_e |C_e|$. Exact marginal work is $O(c \sum_t W_t)$, versus $O(cT|V_\Omega|)$ for full-set greedy or an exact audit along the returned trajectory. Both expressions bound marginal-evaluation work; neither predicts end-to-end wall time.

Online screening versus offline audit. Along the returned trajectory, let $m_t = \Delta(e_t \mid S_{t-1})$, and let m_t^* be the best feasible exact marginal over the complete scenario-filtered candidate set at the same incumbent. Then

$$\delta_t = m_t^* - m_t \geq 0, \qquad C(\Omega) = \sum_t \delta_t \quad (3)$$

Here, δ_t is the exact missed gain at round t, while $C(\Omega)$ accumulates it along the returned trajectory. Computing m_t^* requires full-set exact-marginal scans and is in the same cost class as full-set greedy; we use it offline or at high-stakes checkpoints and exclude it from the reported rollout times. This audit is neither the final coverage gap between two different greedy trajectories nor an optimality certificate.

For nonnegative weighted coverage, $\Delta(e \mid S) \leq f(\{e\})$. CLIPPER can therefore reuse cached singleton scores and scenario masks to compute a conservative residual screening bound

$$u_t = \max_{\substack{e \in V_\Omega \setminus S_{t-1} \\ S_{t-1} \cup \{e\} \in \mathcal{I}(\Omega)}} f(\{e\}), \qquad \hat{\delta}_t = \max\{0, u_t - m_t\}.$$

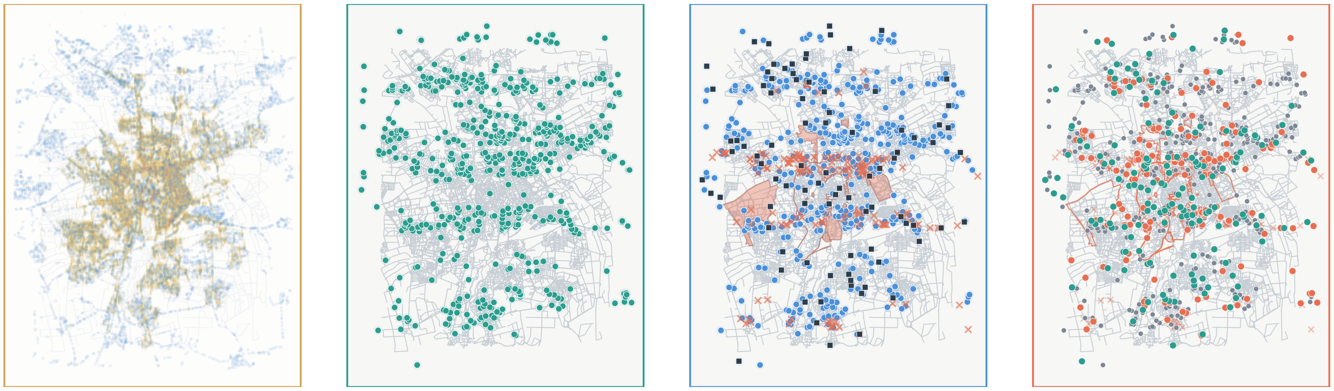
With $u_t = 0$ for an empty feasible residual set, $0 \leq \delta_t \leq \hat{\delta}_t$. This avoids residual exact-marginal recomputation but can still require filtering all residual candidates. CLIPPER therefore uses this conservative bound only to trigger pool expansion or an offline audit, not as a live quality certificate. If a pool has no positive feasible marginal before the budget is filled, expansion or a full scan is required to distinguish true termination from shortlist truncation.

Replay contract. The standard payload records scenario and policy settings, aggregate counters, termination status, and a fingerprint of the selected set; selected IDs are optional, and configured audits add aggregate missed-gain values. Given identical stored inputs and deterministic ranking and tie rules, reruns should reproduce the fingerprint and aggregates. This checks implementation determinism, not data reproducibility or planning validity.

3 Evaluation

Figure 2 shows a separate 25 m Braunschweig diagnostic replay; Table 1 reports means over the complete chains.

Data construction and edits. The benchmark retains every valid record in three non-public operator trip feeds available to the project for internal shared-mobility research. Processed inputs remove provider identifiers. After validation, the feeds contain 243 649 Braunschweig trips (2024-01-01–2024-04-30), 4 958 963 Munich trips (2023-12-18–2024-08-22), and 12 306 947 Berlin trips over the latter interval. We snap each valid origin and destination to its nearest feasible support point and assign one unit of demand to each snapped endpoint; temporal bins are summed for the paper's spatial objective. Candidate grids use OpenStreetMap (OSM)-derived sidewalk, footway, and public-parking support [5] after restricted-area filtering, at candidate-grid resolutions of 10 m in Braunschweig and 25 m in Munich and Berlin. Starting from 16 K-means clusters (seed 42), we split or merge clusters until each proposal group contains 1,750–2,750 candidate points. This produces 29/16/36 synthetic proposal groups over 60 495/32 907/68 922 unique candidates in Braunschweig/Munich/Berlin. These groups define quota accounting and are not administrative districts. Coverage and facility spacing use shortest paths on per-city OSM walk graphs, with a 200 m coverage radius and 600/1200/1600 facility budgets. At $K = 1024$, CLIPPER-F generates at most 29 696/16 384/36 864 proposal entries per round



(a) Demand/support (b) Baseline $E0$ (c) Edited $E5$ (d) $E0$-to-$E5$ change

Figure 2: Braunschweig 25 m CLIPPER-F diagnostic replay with controlled benchmark edits, not recorded municipal actions. At $E5$, the plan excludes 162 candidates (×), locks 90 sites (squares), and enforces 25 m spacing; it retains 443 baseline sites and selects 157 replacements. Panel (d) shows retained, removed, and added sites. Primary results average all 11 full-data states at city-specific candidate-grid resolutions.

Mode	Policy	City	Coverage C/FC (%)	Gap (pp)	Rollout C/FC (s)	Speedup
CLIPPER-F(1024)	balanced	BS	90.1/90.4	0.245	1.762/24.029	13.6×
		MUC	69.7/69.7	0.003	1.493/22.943	15.4×
		BER	59.5/59.5	0.001	1.825/52.685	28.9×
CLIPPER-A(8192)	relaxed	BS	95.0/96.8	1.82	3.219/28.343	8.8×
		MUC	81.1/81.2	0.12	3.360/22.931	6.8×
		BER	69.3/69.5	0.27	4.415/48.826	11.1×

Table 1: Full-data means over $E0 \dots E10$. C/FC denotes CLIPPER and its same-policy full-set greedy control; BS/MUC/BER denote Braunschweig/Munich/Berlin. Coverage is rounded to one decimal, while gap uses unrounded means. CLIPPER-F and CLIPPER-A use different cap policies; every mean includes all 11 states rather than selected anchors.

before filtering and deduplication; CLIPPER-A uses 8192 total slots. No demand or candidate downsampling is used.

We rerun every method from scratch on all states of the constructed $E0 \dots E10$ stress chain. Relative to $E0$, the chain increases the share of high-contribution baseline sites receiving 75 m exclusions from 0 to 17.5%, and the locked baseline share from 0 to 35%.

Edit construction. For each state, synthetic locks form a growing prefix of baseline sites ordered by contribution and spatial dispersion; they do not represent observed infrastructure. Table 2 records the exact schedule: $E5 \dots E10$ ramp network spacing from 25 m to 60 m, and $E8 \dots E10$ add hotspot exclusions.

Results. On an Intel Core Ultra 9 285K, rollout time covers the deterministic optimization after scenario initialization: per-round pool construction, exact marginal evaluation, feasibility checks, and selection. The timer excludes scenario construction, a preliminary proposal-shape check, raw preprocessing, optional screening and audits, and output serialization. Table 1 reports means across all 11 states. CLIPPER-F(1024) stays within 0.245 percentage points of same-policy full-set greedy and reduces rollout time from 22.9–52.7 to 1.49–1.83 seconds. CLIPPER-A(8192) uses 9–15% of the relaxed-cap full-set rollout time with gaps of 1.82/0.12/0.27 points. CLIPPER-A answers a distinct, explicitly coverage-prioritized policy question and is not directly comparable with CLIPPER-F. The tabulated means imply cumulative rollout times for all 11 states in one city of 16.4–20.1 seconds with CLIPPER-F versus 4.2–9.7 minutes with same-policy full-set greedy. This difference quantifies savings across repeated scenario comparisons.

Chain sensitivity and evidence scope. Table 2 reports all 33 paired state–city outcomes so edit sensitivity remains visible. For CLIPPER-F, statewise gaps to same-policy full-set greedy span −0.038–0.377, −0.010–0.016, and −0.001–0.002 percentage points in Braunschweig, Munich, and Berlin. Citywise speedup ranges are 7.4–40.7× (BS), 8.4–49.1× (MUC), and 15.7–84.3× (BER). Speedups are largest before network-spacing constraints activate at $E5$, but remain at least 7.4× after spacing and hotspot edits are combined. Small negative gaps do not establish superiority: restricted and full-set greedy can follow different trajectories, and neither control is an optimum. We evaluate $K = 1024$ as the operating point.

Determinism and audit scope. Exact audits are outside the rollout timer. The optional screen can trigger pool expansion or an offline audit. In a separate full-data diagnostic using the non-public inputs, three reruns of every state and city (99 runs) produced identical coverage, step counts, gain-evaluation counts, termination reasons, and terminal flags within each state. Every run ended with no feasible positive-gain candidate remaining, and none indicated shortlist truncation.

State	Constructed edit schedule: Core (%)	Lock (%)	Hot	Space (m)	Braunschweig: Gap	Speedup	Munich: Gap	Speedup	Berlin: Gap	Speedup
E0	0.0	0.0	0	0	0.377	12.8	-0.010	15.7	0.002	28.7
E1	2.5	0.0	0	0	0.377	40.7	-0.010	47.9	0.002	81.4
E2	5.0	0.0	0	0	0.377	39.5	0.016	49.1	0.002	84.3
E3	5.0	10.0	0	0	0.319	40.7	0.004	46.5	0.002	72.5
E4	7.5	10.0	0	0	0.319	38.4	0.004	44.7	0.002	72.7
E5	7.5	15.0	0	25	0.312	10.4	0.004	11.3	0.000	21.2
E6	10.0	20.0	0	35	0.266	9.2	0.004	10.0	0.000	19.9
E7	12.5	20.0	0	45	0.243	9.4	0.004	10.4	0.000	19.3
E8	12.5	25.0	10	50	0.079	8.4	0.004	9.3	0.000	18.0
E9	15.0	30.0	15	55	-0.038	8.0	0.004	8.7	-0.001	17.0
E10	17.5	35.0	20	60	0.064	7.4	0.004	8.4	-0.001	15.7

Table 2: Complete deterministic chain and CLIPPER-F performance. Core gives the baseline-prefix share assigned 75 m **exclusion zones; Hot counts** 150 m **halos around dense endpoint cells. Gap (pp) and speedup are relative to same-policy full-set greedy; negative gaps reflect different greedy trajectories. Each state is independently recomputed.**

4 Limitations and scope

Limitations. The $E0 \dots E10$ chains are controlled tests of repeated comparison, not observed workshops. Neither the agreement nor the benchmark measures edit frequency, response-time thresholds, or planner utility. The weighted point-coverage model omits congestion, compliance, equity, temporal rebalancing, and physical curb capacity, which would require assignment or occupancy variables. We evaluate one operating point per mode with a static singleton ranking on one workstation. A neighborhood-refreshed ranker remains an open latency–quality comparison, and a held-out city–provider pair is needed to assess transfer. The trip feeds are incomplete market samples and nonredistributable. The retained artifact omits selected-ID fingerprints and proxy scores; independent reconstruction requires processed candidates, scenario records, and fixed OSM snapshots. Network-spacing bounds remain too loose for operational approximation guarantees.

Operational interpretation. CLIPPER separates scenario design from optimization. Planners define each policy state; the optimizer independently returns a constraint-feasible coverage plan over a common demand and candidate basis. Holding that basis fixed isolates each policy edit, while replay records preserve its comparison context. A study with professional planners should measure time to an accepted scenario, alternatives inspected, constraint corrections, and the usefulness of replay and audit records.

5 Conclusion

Across the three cities, CLIPPER-F's complete-chain mean coverage stays within 0.245 percentage points of its same-policy full-set greedy control while achieving 13.6–28.9× mean rollout-time speedups. Under its coverage-prioritized policy, CLIPPER-A uses 9–15% of its same-policy control's rollout time with mean gaps of 1.82/0.12/0.27 percentage points. In both modes, the selector enforces every encoded hard constraint. CLIPPER combines bounded deterministic candidate generation, exact constraint checks during greedy selection, same-policy controls, replay records, and offline missed-gain audits. Within the evaluated model, these results establish CLIPPER as a practical computational foundation for replayable city-scale scenario comparison with exact constraint enforcement. Its execution contract also supports alternative candidate generators without changing scenario semantics.

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